
A 12-to-20Gb/s, 2.93-pJ/bit Jitter-Filtering Retimer with High Input Jitter Tolerance in 28nm CMOS



SAMSUNG

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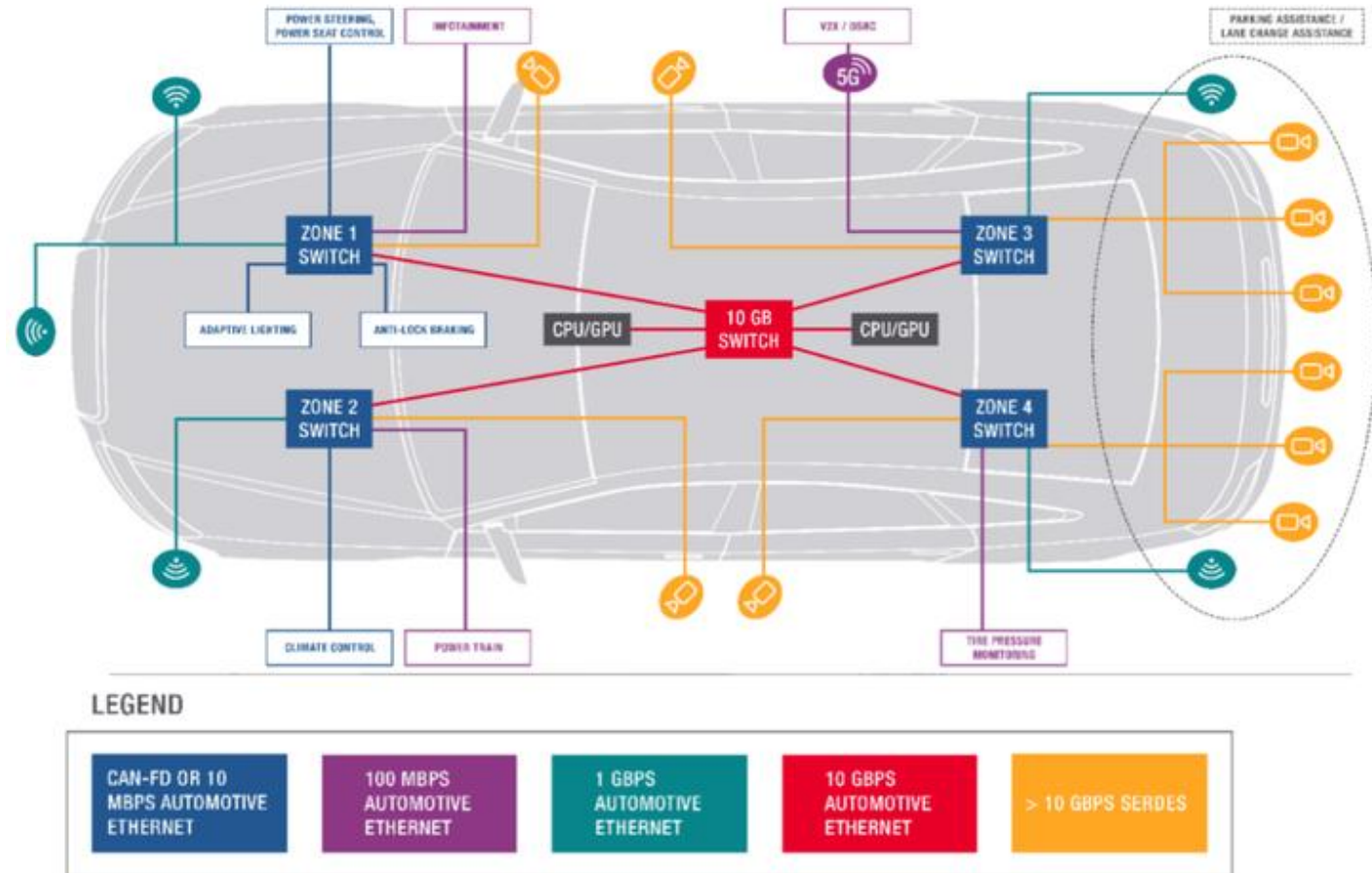
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Outline

- Motivation & Retimer Architecture
- Wide Bandwidth Input CDR
- Jitter-filtering PLL & TX
- Experimental Results
- Conclusion

Motivation

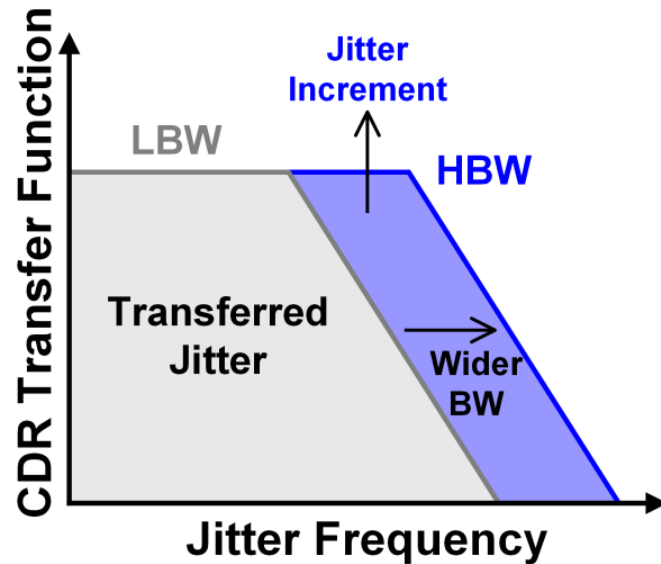
- ❑ Automotive SerDes includes numerous interconnects for networking, sensing and switching components
- ❑ Increasing serial I/O data rates over the channel results in high ISI
- ❑ Jitter-filtering retimers recover the data with high JTOL and transmit to a central unit



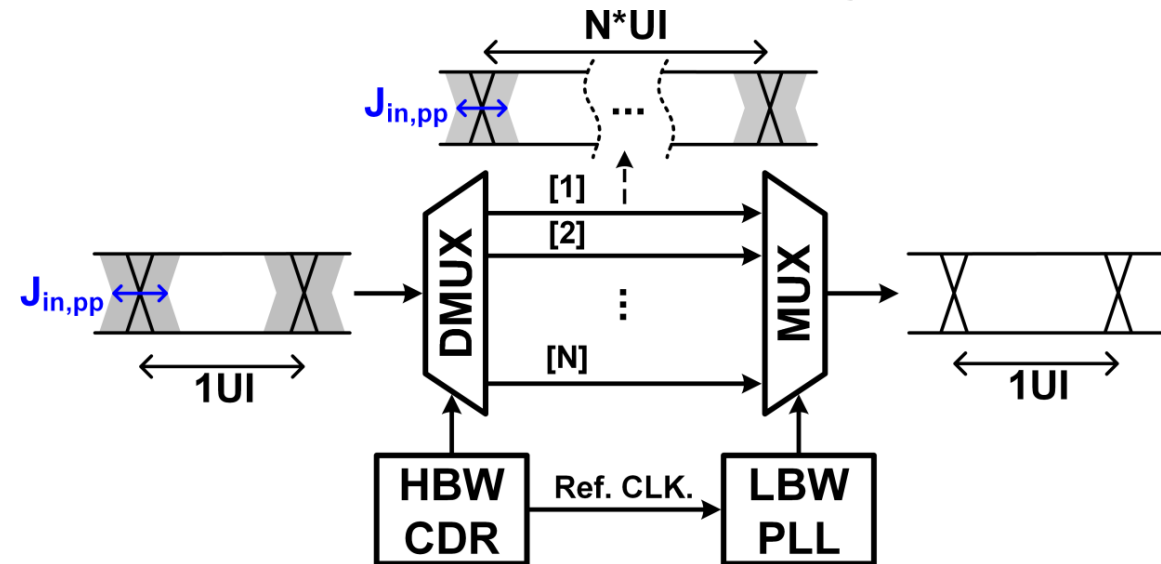
<https://www.keysight.com/blogs/en/tech/educ/2024/automotive-ethernet>

Jitter Filtering Retimer Architecture

CDR BW vs Output Jitter



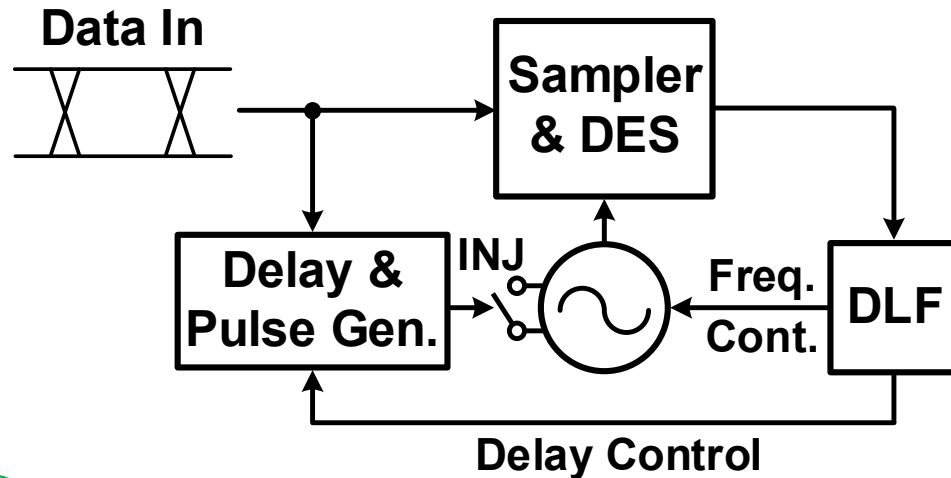
Simplified Block Diagram



- Trade-off in Conventional CDR
 - Wide-BW is required for superior JTOL → More jitter transferred to output
- Proposed Retimer: Cascading Wide-band CDR with Narrow-band PLL
 - Wide-band CDR tracks input jitter faithfully to maximize JTOL
 - Jitter beyond PLL bandwidth is filtered out

CDR Architecture Comparison

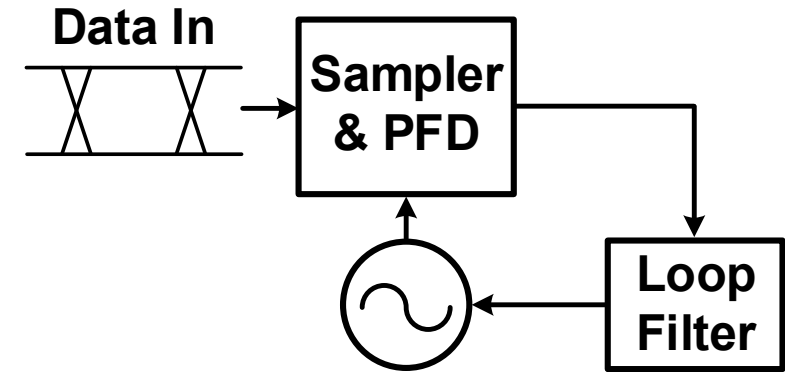
ILO-based CDR



😊 Wide Input Bandwidth and High JTOL

😞 Poor Jitter Filtering

2nd-order PLL-based CDR



😊 Better Jitter Filtering

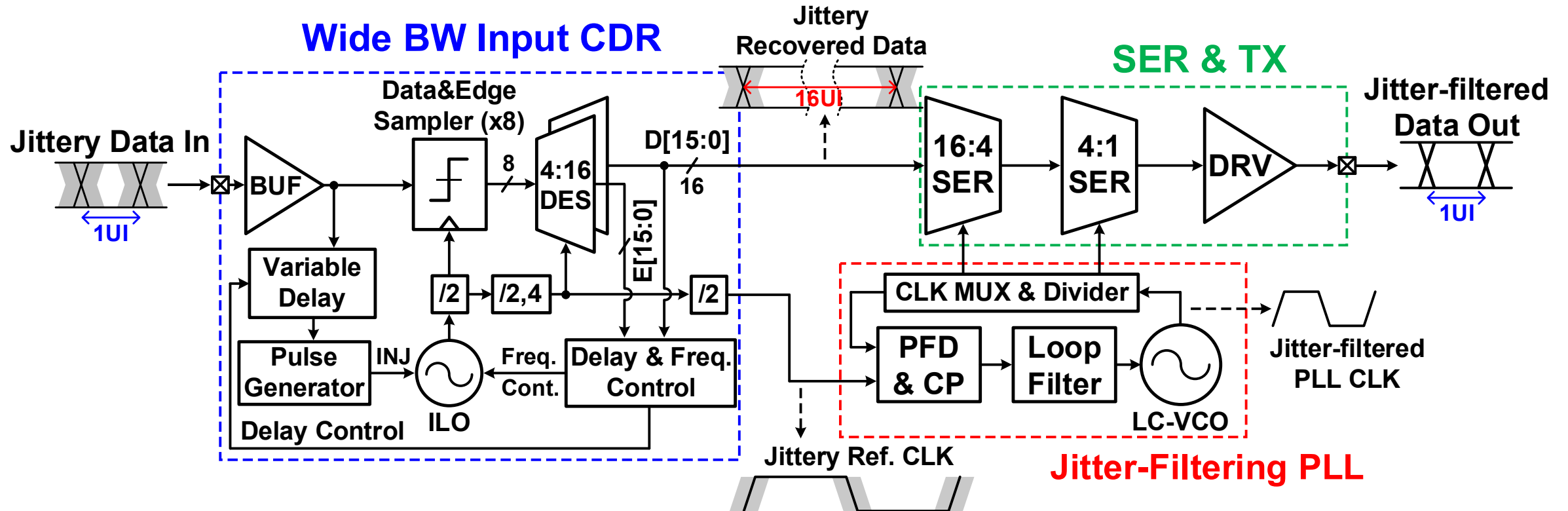
😞 Limited Input Bandwidth and Low JTOL

- ❑ ILO-based CDR offers wide BW since it does not require a loop filter
 - But limited jitter filtering as it tracks input jitter directly
- ❑ This work cascades ILO-based CDR with a 2nd-order PLL, achieving both wide input bandwidth and enhanced jitter filtering

Outline

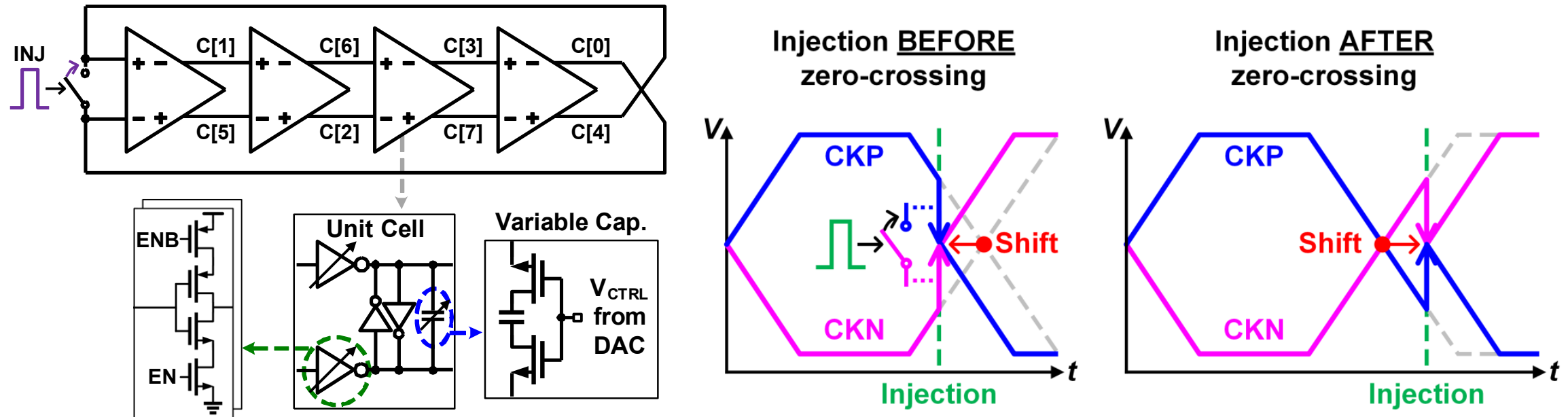
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Retimer Architecture



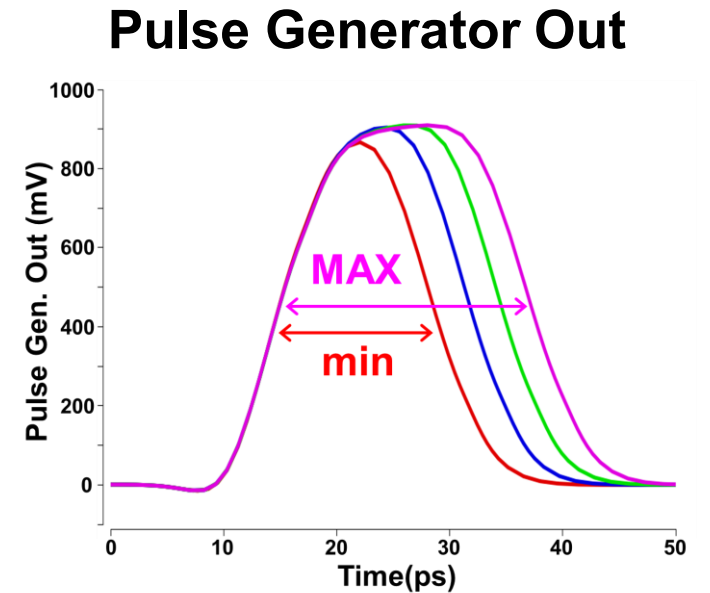
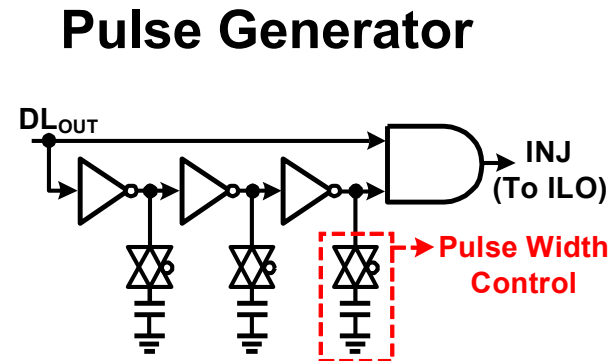
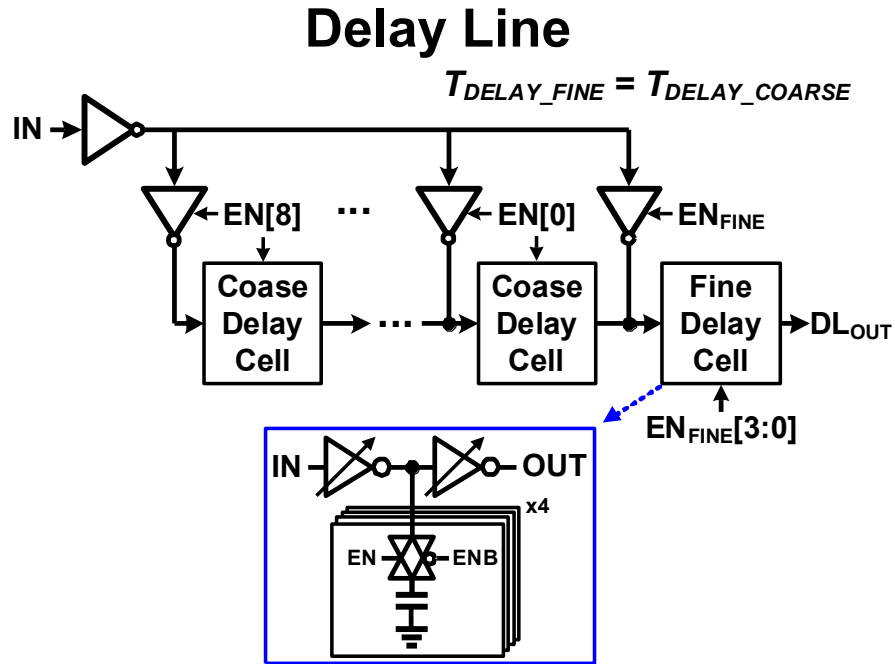
- ❑ Input CDR: ILO-based CDR for wide input bandwidth
- ❑ Jitter filtering PLL with LC-VCO: Narrow bandwidth for jitter-filtering
- ❑ Serializer & TX: Re-serialize the recovered data using jitter-filtered PLL clock

Injection-Locked Oscillator



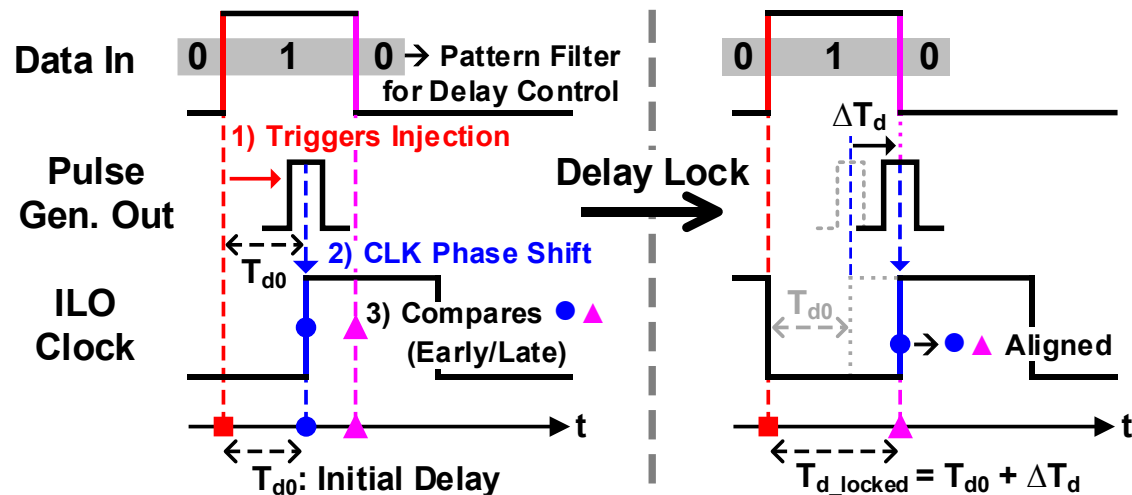
- ILO achieves rapid phase shifting through data edge injection without passing through loop filter
 - Injection switch shorts the differential node, shifting the zero-crossing point toward the injection event

Delay Line & Pulse Generator

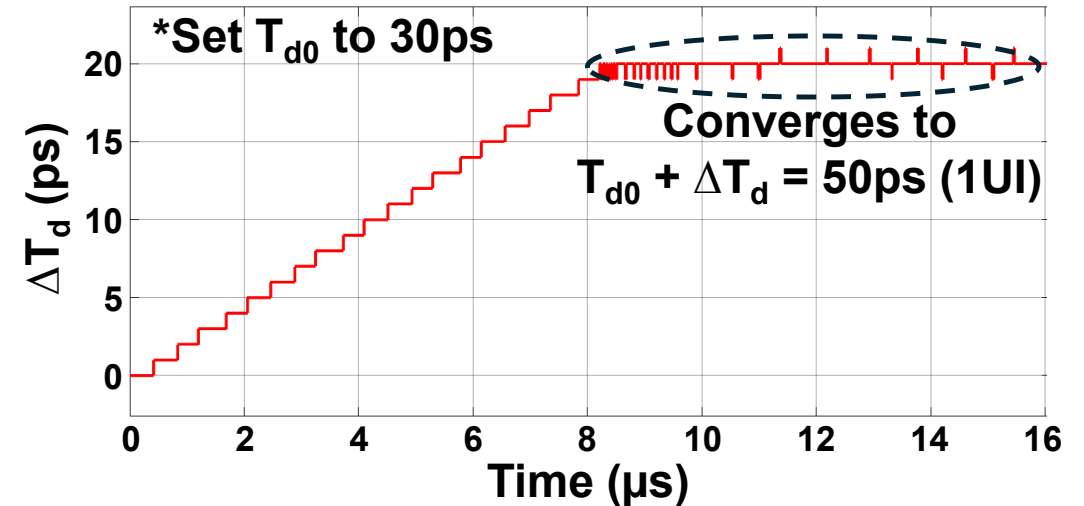


- Inverter-based delay line, rather than logic gates, supports higher data rates of up to 20 Gb/s
- Pulse generator takes the rising edge and generates an injection pulse
 - Switched capacitors modulate the pulse width

Delay Line Control by DLF

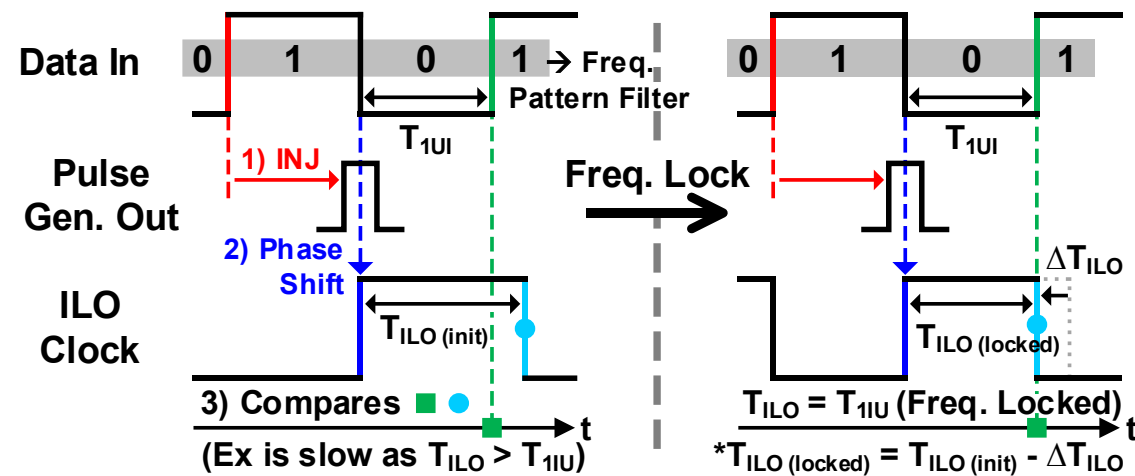


Modeling Simulation of Delay Control

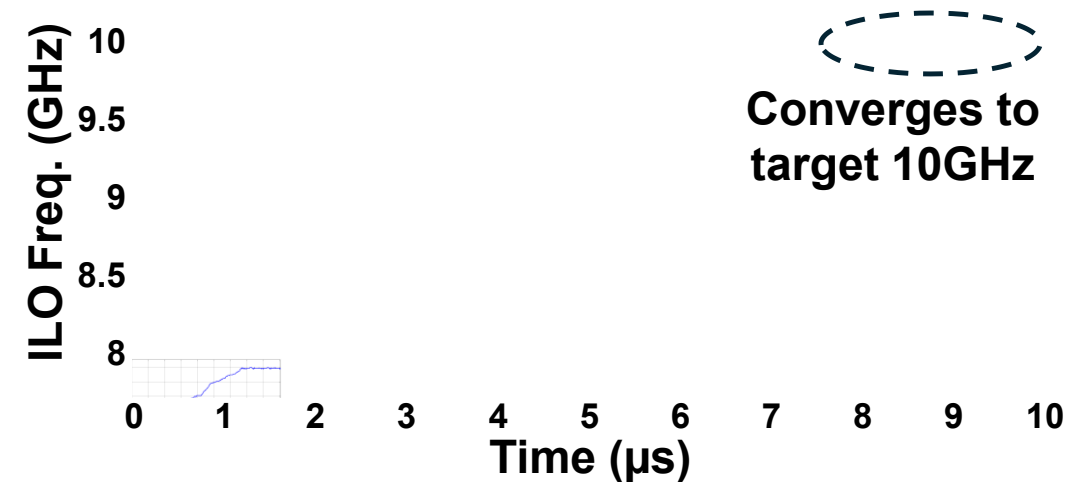


- Delay line control is achieved by comparing the injected ILO clock edge with the data
 - T_{d0} : Minimum delay from data input to ILO injection SW
 - ΔT_d : Additional delay of delay line to align the ILO clock with data edge
- Modeling plots ΔT_d with $T_{d0} = 30\text{ps}$: $T_{d0} + \Delta T_d$ converges to 1UI (50ps)

ILO Frequency Control by DLF



Modeling Simulation of ILO Freq. Control

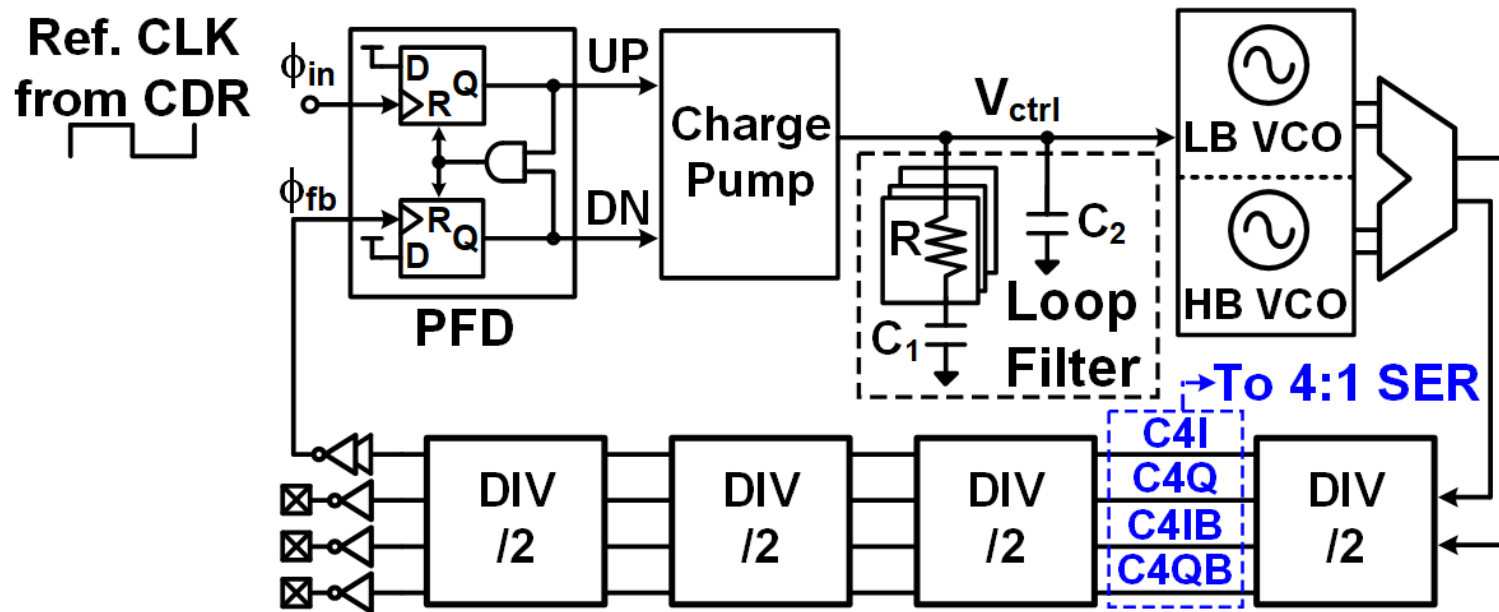


- ILO frequency control is achieved by comparing the ILO clock period with data bit period
- Modeling plots when initial ILO clock frequency is set to 8GHz and converges to 10GHz

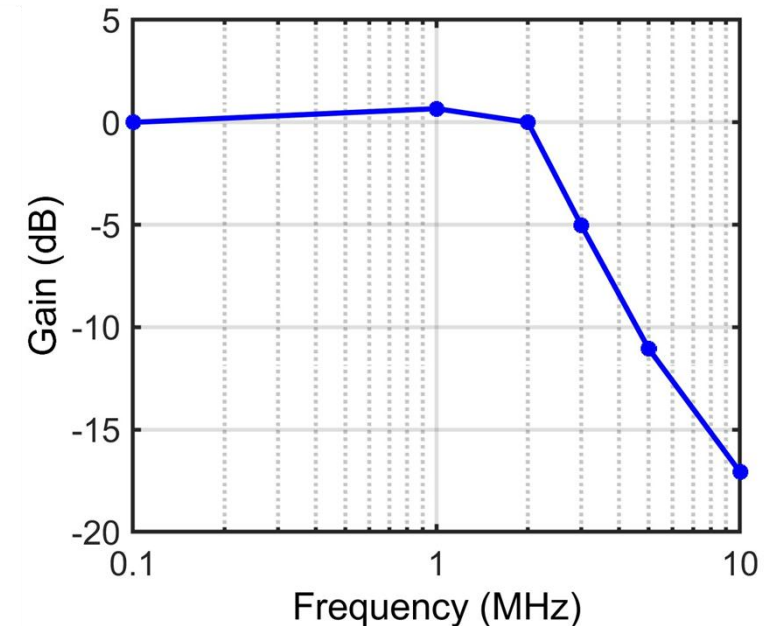
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Jitter-Filtering PLL Architecture



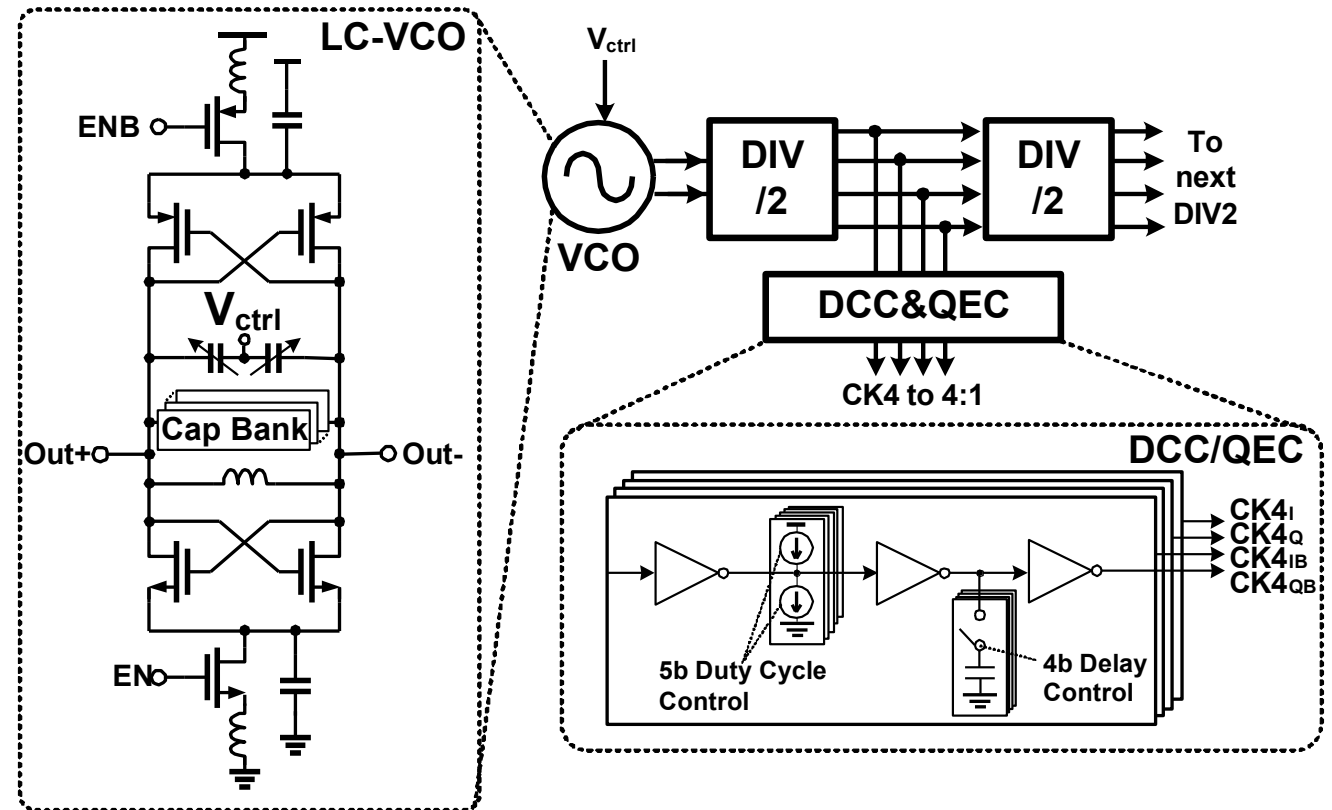
Measured PLL Jitter Transfer Function



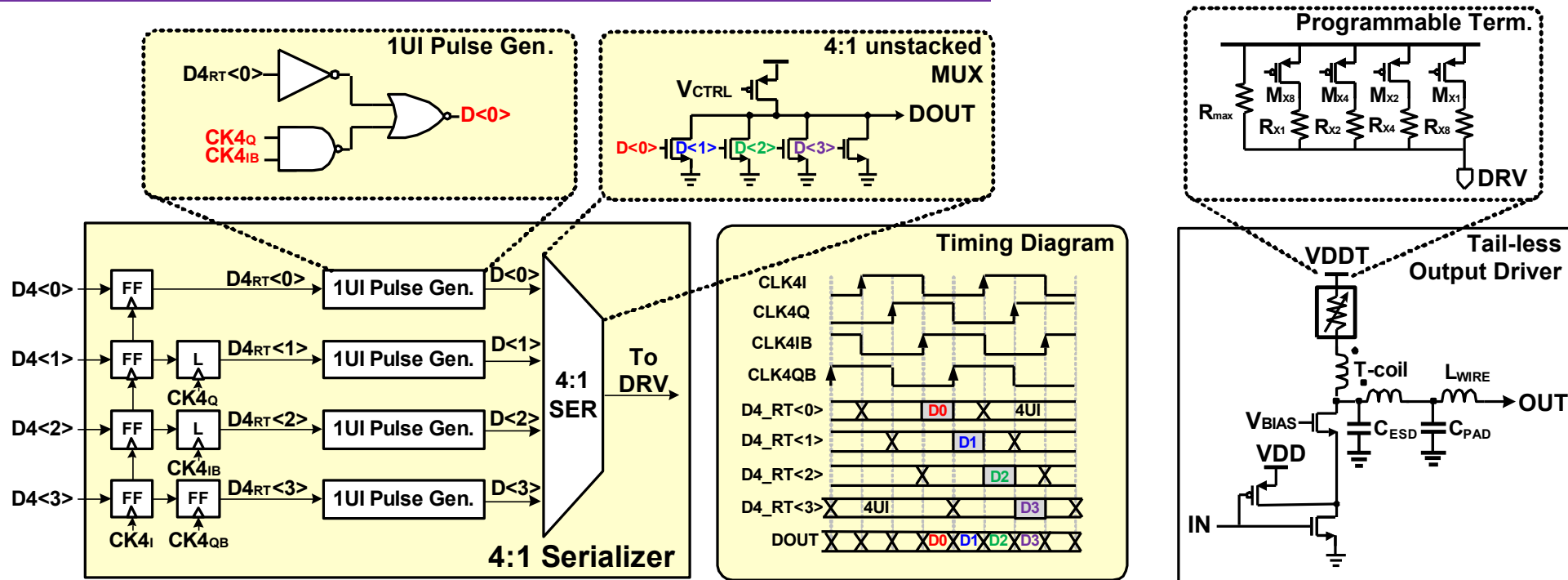
- A low bandwidth PLL filters the high frequency jitter of the recovered clock from the input CDR
- Loop bandwidth and settling time can be optimized by controlling the charge pump current and RDAC in the loop filter

Jitter-Filtering PLL Architecture

- Two LC-VCOs cover the wide range of 6-10GHz clock frequency
- LC-VCO with tail filtering achieves the lower phase noise than ring-VCO
- Quadrature clocks from the divide-by-2 stage are corrected using 5-bit duty-cycle and 4-bit quadrature-error correction circuits

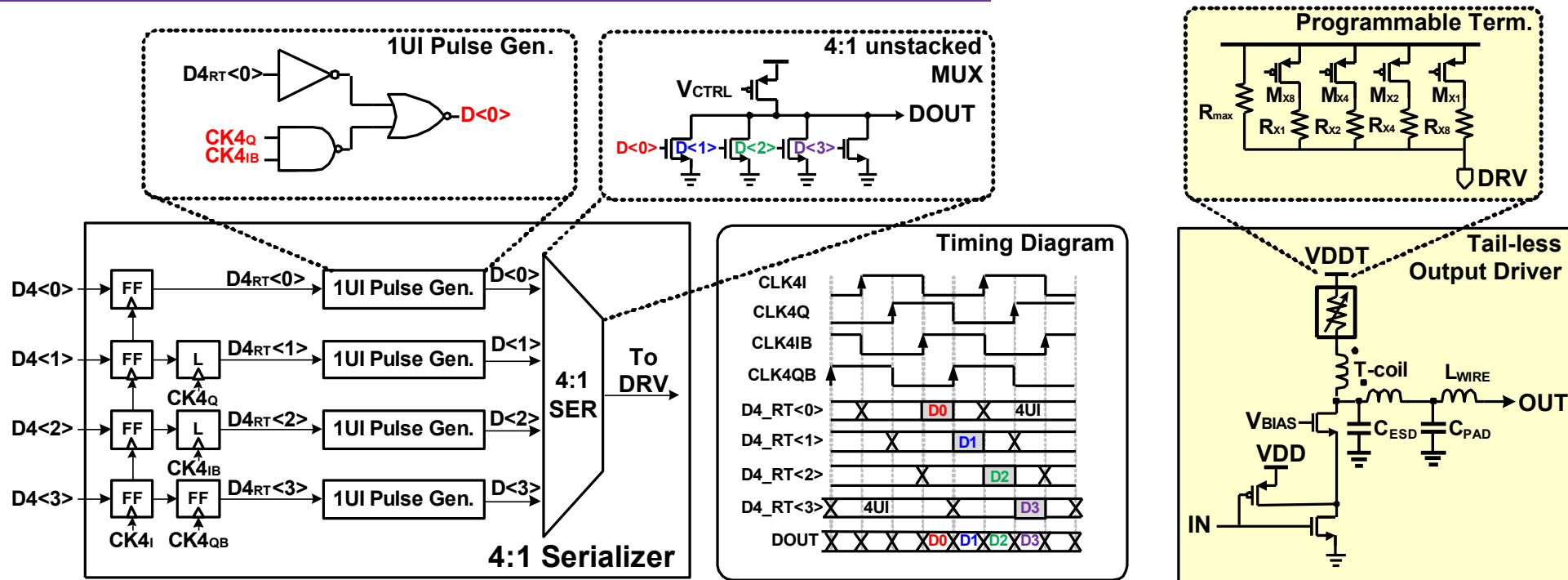


Transmitter Architecture



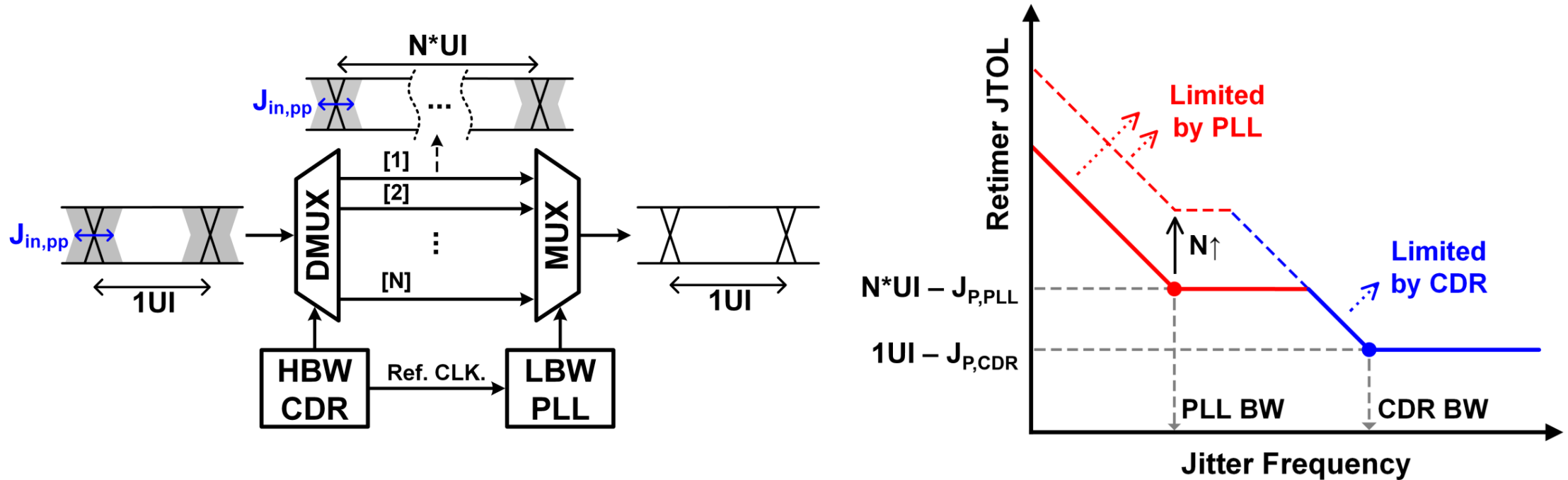
- ❑ Quarter-rate serializer is implemented using unstacked mux which minimizes output bandwidth degradation
- ❑ 4-phase delay circuit carves the 1UI-wide data at the third UI out of the four

Transmitter Architecture



- ❑ Tail-less output driver achieves better bandwidth compared to conventional cascode driver
- ❑ Output network includes 2 kV HBM ESD protection, output driver, package parasitic and T-coil

Retimer JTOL Estimation



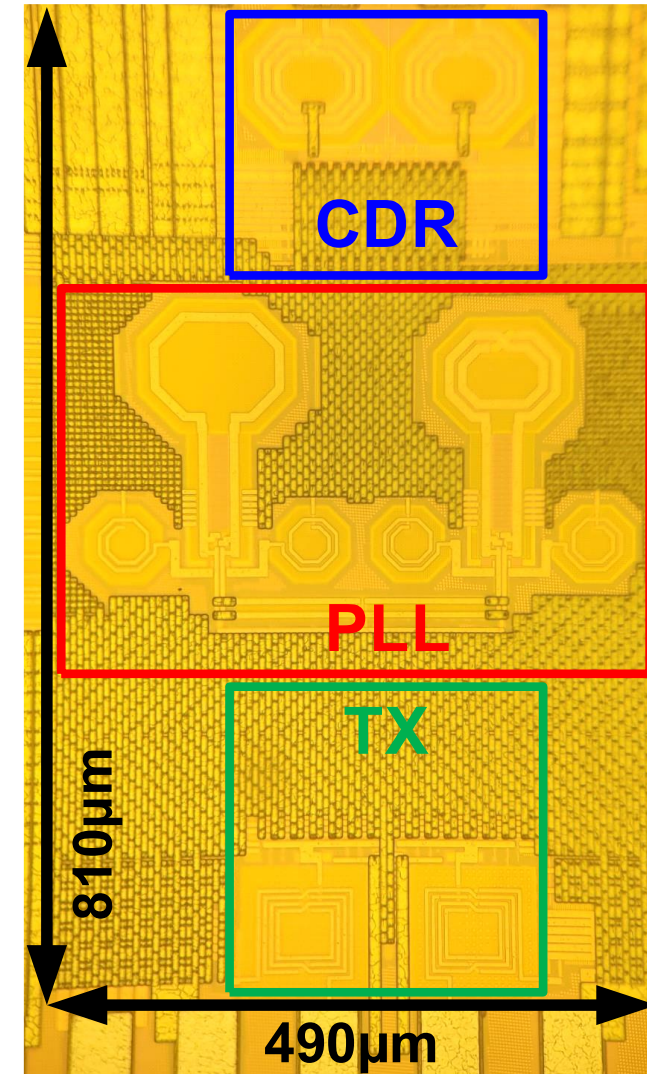
- Due to narrow-BW PLL, deserialization is required to extend timing margin of PLL
- Increasing deserialization factor (N) improves mid-range JTOL
 - But it does not improve high frequency JTOL
 - It also increases latency and system complexity
- Overall JTOL is the min of the CDR JTOL and the PLL JTOL

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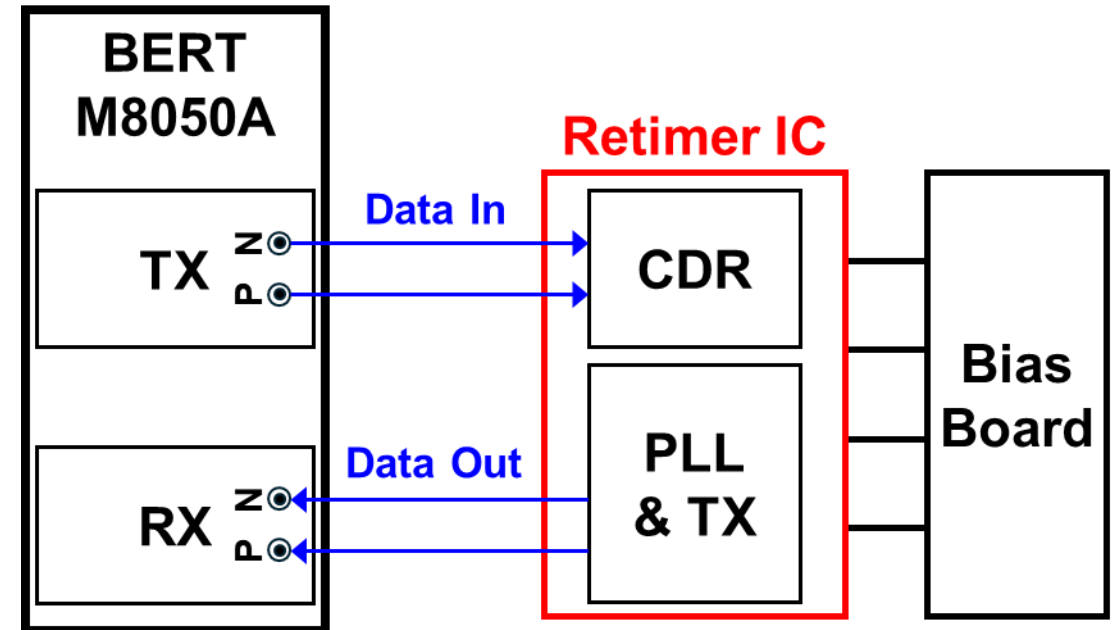
Die Micrograph

- Implemented in 28-nm CMOS
- Area = 0.4mm²
- Two LC-VCOs occupying the majority of the area



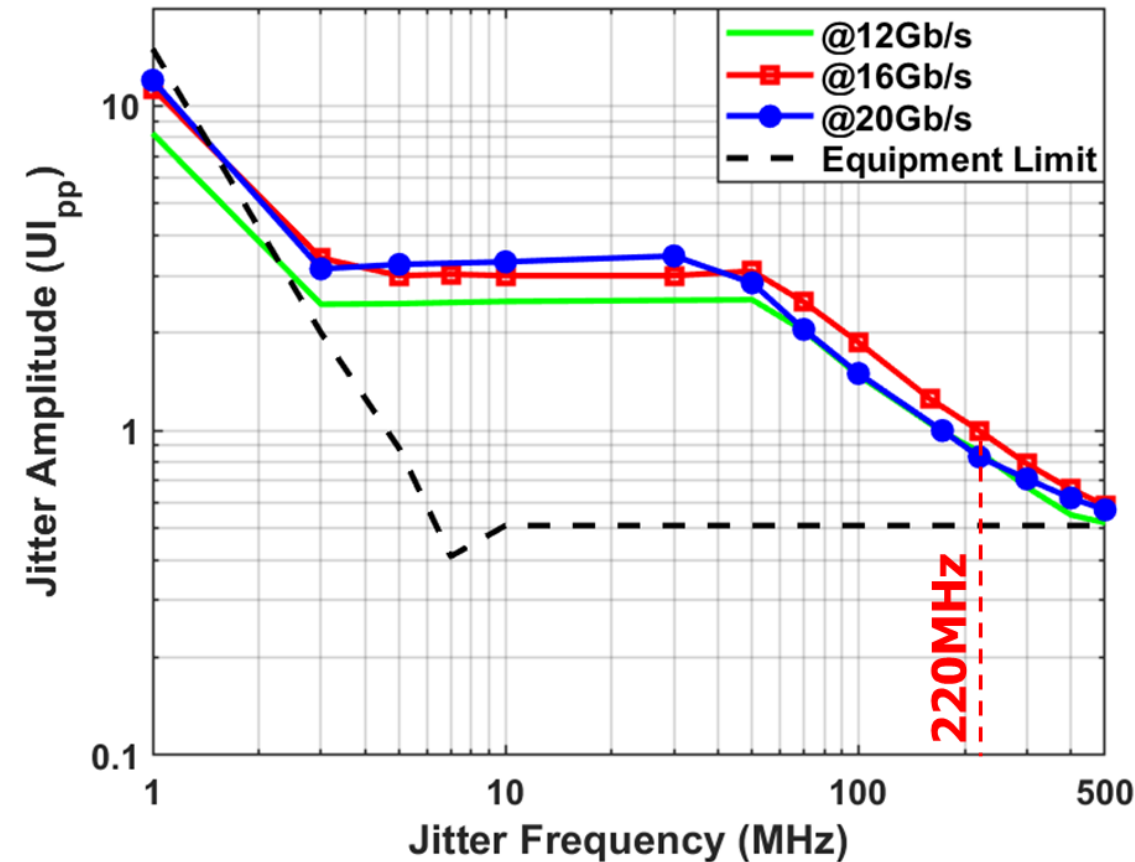
Measurement Setup

- Keysight BERT M8050A
 - BERT TX sends data with injected sinusoidal jitter to the retimer IC
 - BERT RX performs BER calculation to verify retimer IC performance
- Used 0.9V supply voltage

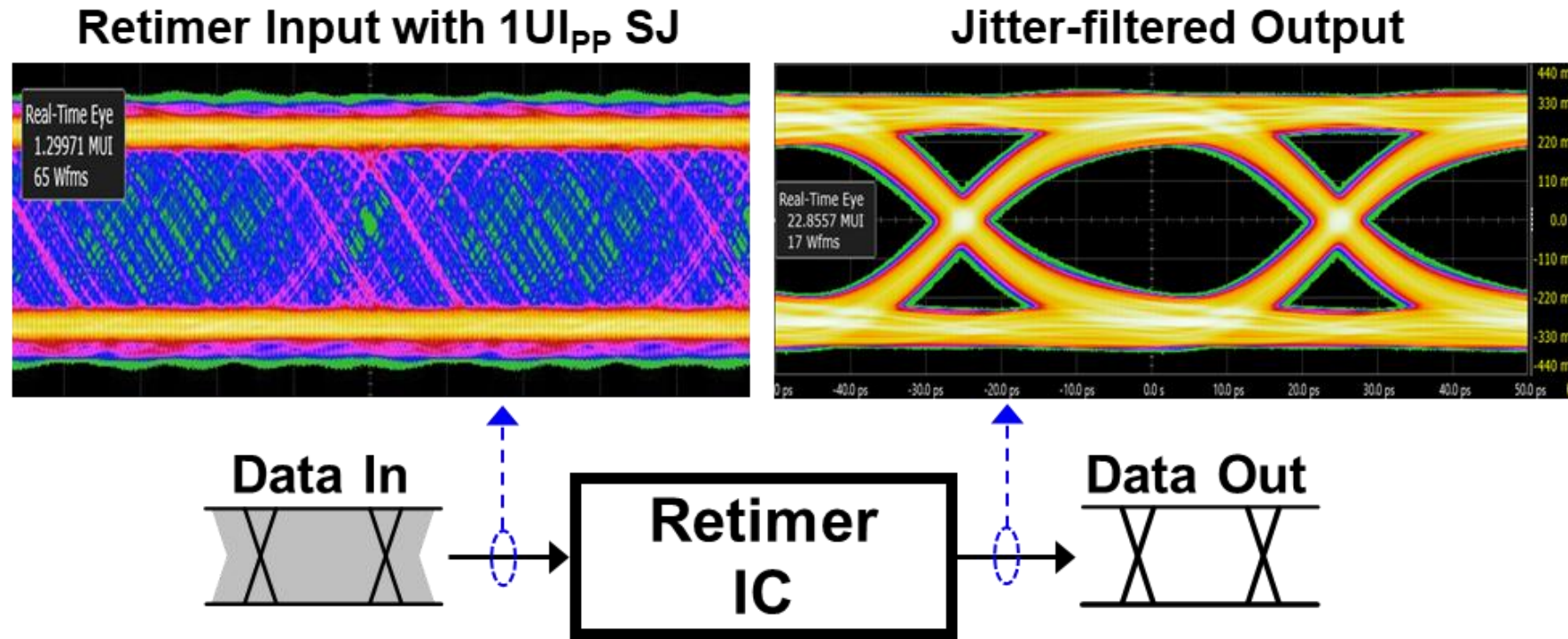


Jitter Tolerance

- 1UI_{pp} JTOL @173MHz (20Gb/s) and @220MHz (16Gb/s)
- $0.57\text{UI}_{\text{pp}}$ JTOL @500MHz (20Gb/s)
- Input CDR BW is estimated to exceed 500MHz
 - Equipment (BERT) limit of 500MHz
- Dashed line represents without the retimer IC connected
- Measurements show good agreement with phase-domain estimation



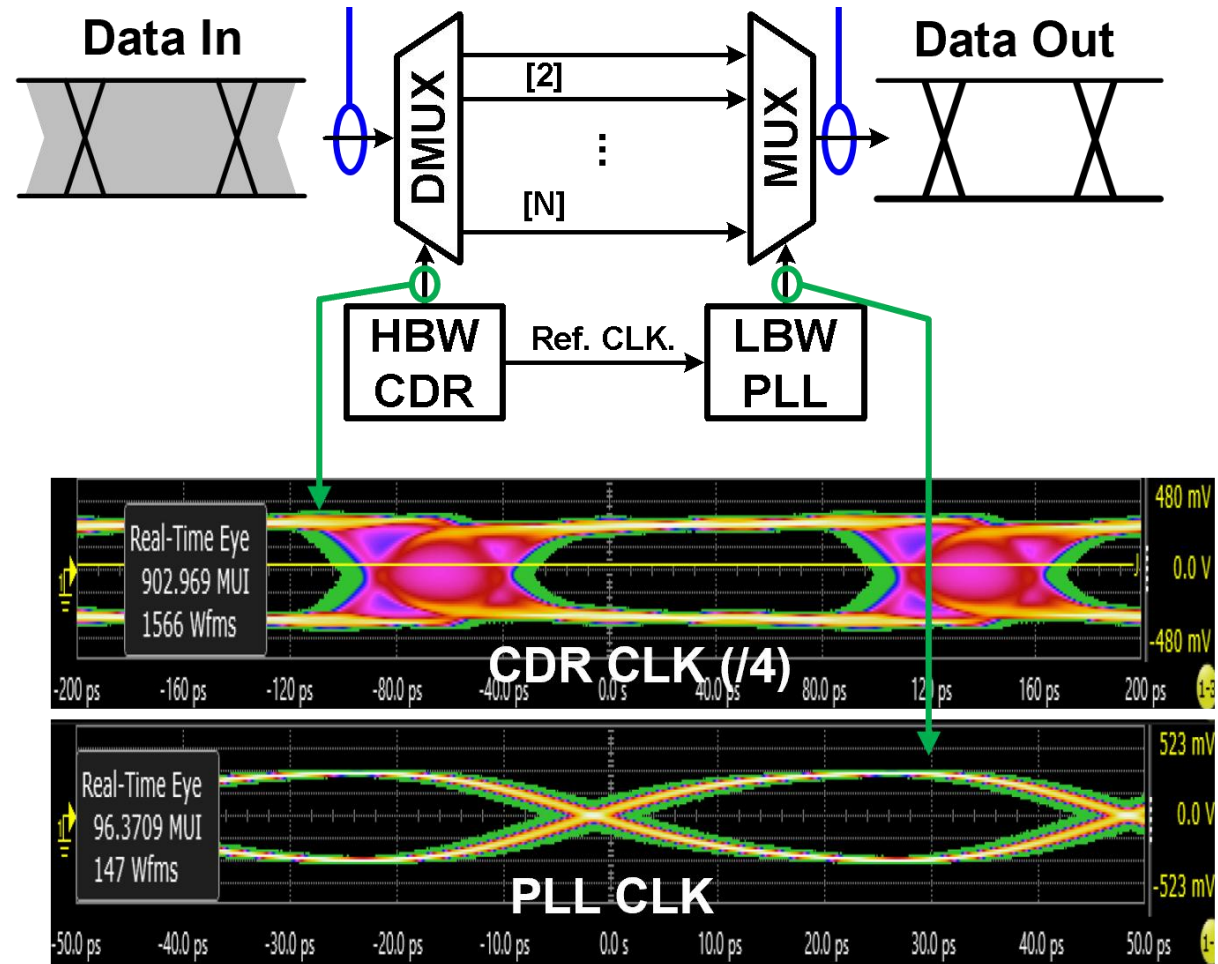
Eye Diagram Measurement



- ❑ 20Gb/s eye diagram with 1UI_{pp} SJ added at 173MHz
- ❑ Closed input eye diagram due to 1UI_{pp} SJ amplitude
- ❑ Wide-open output eye diagram by the jitter-filtering retimer

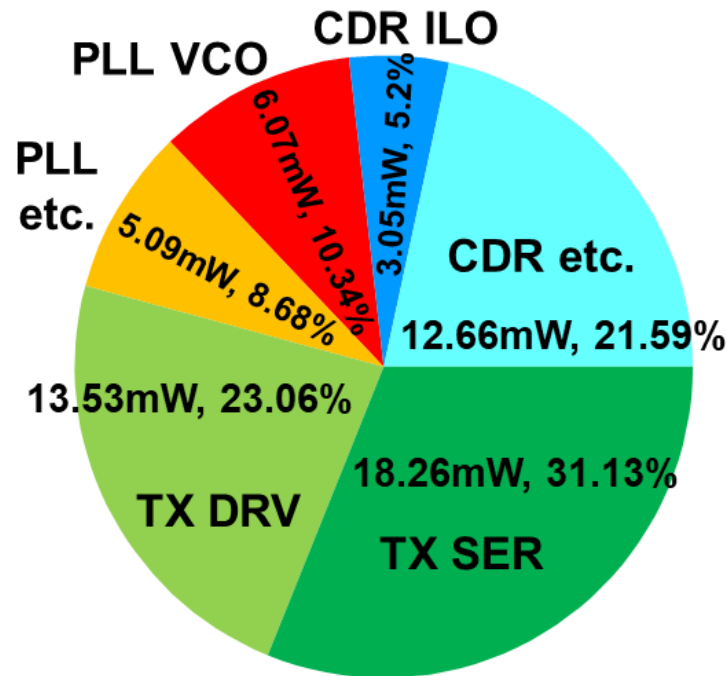
Clock Eye Diagram Measurement

- CDR clock has ~ 1 UI eye closure due to input jitter
- Jitter-filtered PLL clock



Power Breakdown

Total Power = 58.66mW (20Gb/s)
Energy Efficiency = 2.93pJ/bit



- Quarter-rate CDR lowers sampler/deserializer speed than prior half-rate designs, leading to better energy efficiency

Performance Comparison

		ISSCC'08 [6]	JSSC'16 [4]	JSSC'19 [3]	This Work
Technology		250nm	28nm	28nm	28nm
Supply (V)		3.3/1.8	0.9	0.9	0.9
Data Rate (Gb/s)		10.3125	1 – 12	10	12 – 20
Architecture		Full-rate	Half-rate	Half-rate	Quarter-rate
BER		$< 1 \times 10^{-12}$	$< 1 \times 10^{-9}$	$< 1 \times 10^{-12}$	$< 1 \times 10^{-12}$
JTOL	1UI _{pp}	@20MHz ¹	@120MHz ²	@31MHz	@220³ / 173⁴MHz
	@300MHz	₅	0.56UI _{pp} ²	0.2UI _{pp}	0.79³ / 0.71⁴UI_{pp}
	@500MHz	₅	₅	₅	0.59³ / 0.57⁴UI_{pp}
Jitter Filtering?		No	No	No	Yes
Energy Efficiency (pJ/bit)		83.01 ⁶	1.9 ^{2,6}	1.28 ⁶	0.79^{4,6} / 2.93^{4,7}

¹Estimated from JTOL measurements

²At 12Gb/s, Measured the BER using an internal error checker

³At 16Gb/s

⁴At 20Gb/s

⁵Not reported

⁶CDR and receiver part only (Does not include jitter-filtering PLL and TX)

⁷Includes CDR, PLL, and TX

Conclusion

- Proposed jitter-filtering retimer achieves both wide input bandwidth and strong jitter filtering — overcoming a common trade-off in conventional designs
- ILO-based input CDR enables fast tracking of high-frequency input jitter, while a low-bandwidth PLL filters unwanted jitter components that input CDR tracks
- Achieved 20Gb/s operation with JTOL BW > 500 MHz, and JTOL of 1UI_{pp} at 173 MHz with an energy efficiency of 2.93pJ/bit

Acknowledgements

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